SwarmAlpha.

*The first open-source **cognitive governance runtime** for LLM multi-agent systems.*

Security governance prevents agents from **doing harm**. Cognitive governance prevents agents from **thinking wrong**. The agent governance stack needs both. SwarmAlpha handles the second — detecting echo chambers, authority bias, polarization, and premature consensus during LLM agent collaboration.

THE PROBLEM

- **Premature consensus** — agreement in round 1; critical info never discussed.
- **Authority bias** — one overconfident agent dominates the rest.
- **Echo chamber** — similar agents confirm each other's blind spots.
- **Group polarization** — divergence hardens into deadlock.
- **No existing framework** detects or intervenes on these failures.

WHAT WE BUILT

- **Embeddable governance runtime**, not a framework — drop-in layer for any MAS.
- **3 value dimensions**: process monitoring, decision audit, adaptive intervention.
- **LLM perception / Math evolution** separation — agents extract beliefs, math does the rest.
- **Framework-agnostic SDK** — ~2-4h integration into AutoGen / CrewAI / LangGraph.
- **13,000+ lines** of TypeScript, 219 automated tests, full open source.

Key Research Contribution — Diagnosing the Governance Loop

We identified **4 root cognitive defects (D1–D4)** in the prevailing MAS discussion paradigm — missing state awareness, no per-agent history, synchronous scripted turns, and fabricated influence networks. All 4 were fixed on 2026-07-12, and the experiments were re-run on Crisis (2026-07-14) with the loop closed — confirming governance effectiveness (d=0.84, τ +48%) that prior "governance is ineffective" conclusions had masked.

EXPERIMENTAL EVIDENCE — 89 RUNS ACROSS 2 TASKS (CRISIS 45 + SUPPLIER 44)

Task	Baseline τ	Full Governance τ	Shuffle Control τ	Cohen's d (full vs none)
Crisis (3 rounds, n=15/cell)	0.387 ± 0.160	**0.573 ± 0.271**	**0.760 ± 0.241**	**+0.84**
Supplier (3 rounds, n=15/15/14)	0.680 ± 0.211	**0.787 ± 0.177**	0.671 ± 0.202	**+0.55**

✅ **Governance direction consistent** across both tasks — full > none replicates.

✅ **"False consensus" replicates** — consensus-quality r ≈ 0 in both tasks.

⚠️ **Shuffle has boundary conditions** — effective on hard tasks, noise on easy ones.

🧪 **Bayesian reanalysis**: P(d > 0) = 96.7%, 95% HDI [−0.04, 1.13].

贺孟元 (Mengyuan He) — Grade 10, independent architecture & implementation
TypeScript · Next.js · DeepSeek API · Vitest · 219 tests · 89 experiments · 2 tasks validated

github.com/mulasakee17/swarmalpha
AI-assisted coding · architecture & experiments autonomous